Ekansh Chawla

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Education

Simon Fraser University Expected September 2027
B.Sc. in Computing Science GPA: 3.88

Experience

Amazon Web Services May–September 2026
Software Development Engineer Intern

- Built a Java cross-stack reference detection and generation engine for CloudFormation, reducing multi-day stack reorganization workflows to a single API call.
- Implemented directed-graph dependency resolution with cycle detection, recursive AST traversal, and context-aware generation for five CloudFormation intrinsic function types.
- Engineered atomic resource-ownership transfers with DynamoDB transactions, SWF orchestration, and idempotent retries for zero-downtime infrastructure migrations.

Visier January–April 2026
Full-Stack Developer Intern

- Optimized concurrent analytics pipelines using Scala, Akka Streams, and Futures; reduced API latency by more than 30% through batching and caching.
- Built Angular/RxJS dashboards for real-time analytics and integrated LangSmith-based agent workflows to support nontechnical insight discovery.

Amazon May–September 2025
Software Development Engineer Intern

- Built an SPI-based payment-preference plugin architecture, reducing preference-related database calls by 15%.
- Improved concurrent state management with thread-safe singletons and lazy initialization; reduced contention by 35% in 500-thread JMH benchmarks.
- Maintained more than 100 unit tests and 40 integration tests using JUnit and Mockito in CI/CD workflows.

Research

Fewer Rebalances, Lower Costs? Target Dynamics and Portfolio Implementation
Independent working paper; Zenodo, October 2026

- Built a Python research framework comparing calendar and deviation-based portfolio rebalancing with lagged execution, self-financing accounting, transaction costs, and paired block-bootstrap inference.
- Found 48.5% fewer rebalance events but only 2.7% less traded notional in the primary 2015–2025 comparison; quantified tracking error and robustness with paired block-bootstrap confidence intervals.

Selected Projects

Mercury *C++20, CMake*

- Implemented a price-time-priority order book and state-dependent stochastic simulator with seeded randomness, a virtual clock, configurable order flow, and event-level recording.
- Added invariant and property tests, simulator reproducibility and parameter-response checks, a submission-latency benchmark, and sanitizer build support.

NodeAI *TypeScript, React, WebSockets, MQTT*

- Built a fleet-orchestration prototype with schema-validated mission plans, grid pathfinding, telemetry adapters, and policy-based reassignment for blocked or low-battery robots.

Technical Skills

Languages: C++, Python, Java, Scala, TypeScript, SQL
Systems: Distributed systems, concurrency, graph algorithms, event-driven simulation, performance profiling
Tools: AWS, CloudFormation, DynamoDB, Akka Streams, CMake, JMH, JUnit, Git, Docker